Intelligent Suspect Profile Detection Using Random Forest and Behavioral Features

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***Abstract*—Criminal investigations often depend on the manual interpretation of crime patterns, victim characteristics, and behavioral cues to pinpoint potential suspects, and are hence time-consuming and subject to human bias. This research proposes an automated Suspect Profile Detector that is a machine-learning–based framework for predicting key attributes of suspects based on features related to the crime, such as Modus Operandi, victim demographics, and crime circumstances. The proposal uses a multi-output classification model based on Random Forest that can run predictions on suspect age group, gender, and other relevant profiling parameters after its training using curated datasets pertaining to crime. Extensive model evaluation using accuracy, precision, recall, and confusion matrix analyses facilitates verification about the model's ability to identify consistent behavioral correlations and produce reliable suspect profiles. Therefore, this approach enhances efficiency in investigation and decreases the dependency on subjective judgment to offer a scalable solution for data-driven criminal profiling. This is an indication of how artificial intelligence can work as a supportive tool in modern law enforcement and forensic analysis.**

***Index Terms*—Suspect profiling, Machine learning, Random Forest, Crime prediction, Modus Operandi, Criminal investigation, Behavioral patterns, Crime analytics.**

I. Introduction

Criminal profiling is usually done by forensic psychologists and investigators who look at behavioral evidence to determine offender traits. While this approach works, it has limitations, such as being subjective, hard to scale, and struggling with large amounts of crime data. As digital crime records become more complex, there is a greater need for systematic, data-driven tools.

Machine learning provides effective ways to find patterns that traditional analysis might miss. Rather than reaching absolute conclusions, computational models can identify behavioral links, group similar incidents, and reveal trends that improve decision-making in investigations. This research develops a computational suspect-profiling system that uses anonymized crime datasets to predict possible offender traits through a multi-output machine-learning framework.

The goal is to combine understandability, fairness, and responsible design while showing how machine learning can support expert judgment instead of replacing it in criminal investigations.

II. Literature Review

A. Traditional Offender Profiling

Conventional criminal profiling techniques took their roots from forensic psychology and criminology, with investigators drawing offender traits by analyzing crime-scene behavior, victim selection, location choices, and emotional patterns. The qualitative methods that were previously practiced by the FBI's Behavioral Science Unit, among others, are very much dependent on the expertise and the experience of the case. Although they are very effective in particular situations, traditional profiling is still plagued by a number of major limitations such as subjectivity, lack of empirical validation, inconsistency among profilers, and inability to handle large volumes of data. These constraints point to the necessity for data-driven tools that are systematic and have replicable results.

B. Evolution of Data-Driven Crime Analysis

The growing number of digital crime records has led to a complete transformation of the research area into quantitative analysis that can uncover behavioral and situational patterns. The first computer-based models used among other techniques logistic regression, association-rule mining, clustering, and geographic hotspot analysis – all of them statistical methods. On the one hand, these methods allowed for the systematic detection of crime trends, but on the other hand, they faced challenges with real-world datasets characterized by categorical values, non-linear relationships, and multiple interdependent labels (e.g., suspect age group, gender, race). Further-

more, traditional statistical models are often constrained by the necessity to meet stringent requirements regarding the underlying data distributions, thereby limiting their use in studies of complex criminal datasets.

C. Machine Learning for Offender Behavior Prediction

Machine learning has become a very powerful option for crime analytics owing to its capability of recognizing patterns in noisy, heterogeneous, and high-dimensional datasets. Several algorithms like Support Vector Machines, Random Forests, Gradient Boosting, and Neural Networks have been utilized in various crime-related tasks, including the prediction of crime types, forecasting of crime hotspots, assessing risks, and classifying behaviors. Random Forests, among other methods, are extensively utilized in criminology studies due to their ability to process categorical data, their robustness against overfitting, and their capability to represent non-linear relationships between the features of crime type, victim's characteristics, and the method used. These attributes make Random Forests the most appropriate algorithm for predicting suspect characteristics.

D. Multi-Label and Multi-Output Prediction in Criminology

Investigative situations in the real world generally necessitate the simultaneous prediction of various offender characteristics such as gender, age group, and race. Multi-output learning frameworks distinguish themselves from conventional single-target classifiers by taking into consideration the interdependencies of the output variables and by treating them as a whole when modelling. The MultiOutputClassifier technique encloses a basic estimator like Random Forest and, consequently, the training of all sub-models—one for each target attribute—is performed simultaneously under the same input features. Previous studies have found that multi-output learning enhances the reliability of the predictions, decreases the amount of noise around the predictions, and is more capable of showing the intricate behavioral relationships among the traits compared to the case of having independent models trained separately.

E. Role of Behavioral and Situational Features

For quite some time now, victim characteristics, crime types and methods of operation have been considered significant factors indicating criminal behavior in criminology literature. Research on behavioral linkage and situational crime analysis points to a strong correlation between the mode of a crime, the victim's characteristics, and the perpetrator's social background. The use of these features in machine learning models is advantageous as they represent the basic behavioral patterns that common algorithms might not recognize.

III. METHODOLOGY

A. Dataset Acquisition and Preparation

The system makes use of an anonymised crime dataset that includes suspect characteristics, criminal categories, victim demographics, and methods of operation. The dataset goes through a number of validation procedures to guarantee dependability prior to modeling, elimination of incomplete records with conflicting or missing values.

B. Data Preprocessing

A multi-stage preparation pipeline is used because the dataset includes both numerical and category variables.

1) Categorical Variable Encoding

One-Hot Encoding is used to convert categorical features—such as victim gender, criminal type, and method of operation—into numerical format. As a result, non-numeric data can be interpreted by the Random Forest classifier without requiring ordinal relationships between categories.

2) Numerical Variable Scaling

To increase consistency and lessen the impact of scale changes across characteristics, victim age—the only continuous input—is standardized using z-score scaling.

3) Train–Test Split

To assess generalization performance, the preprocessed dataset is split into training and testing subsets. The split guarantees that the model is verified with data that hasn't been seen before.

C. Feature Engineering

The goal of feature engineering is to improve prediction accuracy by choosing and modifying pertinent variables. Important characteristics for suspect profiling consist of:

- Age of the Victim
- Gender of the Victim
- Type of Crime
- Modus Operandi (MO)

The criminological literature, which emphasizes the substantial correlations between victim traits, MO patterns, and perpetrator demographics, provides the basis for the selection of these features. Their applicability is further confirmed by feature-importance analysis during training.

IV. MODEL ARCHITECTURE

This study employs a RandomForestClassifier wrapped inside a MultiOutputClassifier to perform simultaneous prediction of multiple offender attributes.

A. Multi-Output Classification Framework

Instead of training three separate models, the job is treated as a multi-output classification issue because in real-world investigations necessitate predicting multiple suspect attributes together. For every target label (gender, age group, race), the Multi-Output-Classifier builds internal Random Forest estimators and trains them concurrently using shared input features.

B. Random Forest Estimator

The Random Forest method was chosen because of its:
resilience to overfitting,
capacity to manage category data with large dimensions,
interpretability using feature-importance ratings, and
outstanding results on datasets that are unbalanced.

To reduce bias in favor of majority classes, balanced class weights are used.

C. Model Training and Prediction Workflow

The processed training subset is used to train the model. While undergoing training:

Using bootstrapped samples, each Random Forest sub-model builds several decision trees.

The split criterion is either entropy or Gini impurity.

To minimize variance, predictions are aggregated by majority vote.

Before producing suspect-profile predictions, the same preprocessing pipeline is used for fresh user-provided inputs to guarantee consistency.

V. Performance evaluation metrics

In order to have a clear picture of our machine learning model's performance, we rely mainly on a defined set of evaluation metrics. These metrics allow us to evaluate the system's ability to predict the main suspect characteristics with precision, which include gender, age group, and race, the crime incident details being the basis of the prediction. They also indicate the model's error patterns, for instance, mixing up age groups or wrongfully attributing a suspect's gender or race. Through the study of these metrics, we are able to see the complete view of the model's capabilities and the points where it might still be making mistakes.

A. Accuracy

This parameter indicates the overall correctness of the model's predictions. It is computed by the number of correct predictions (both true positives and true negatives) divided by the total number of predictions. Although accuracy gives a rough idea of the model's performance, it may not be very reliable in case of imbalanced datasets, where one class is dominant over the other substantially. "' Accuracy = (True Positives + True Negatives) / (Total Predictions) "'

B. Precision

In our suspect profiling model precision merely indicates how frequent the system's positive predictions regarding a certain suspect attribute—like gender, age group, or race—are indeed correct. It is obtained by dividing the true positive predictions by the total number of cases the model predicted as positive (true positives + false positives). A high precision score indicates that the model hardly ever mistakes one attribute for another and is good at not producing false positives.

Precision = True Positives / (True Positives + False Positives)

C. Recall

Recall describes the ability of the model to detect positively any case that was present True Positive and False Negative. Put differently; it tells the number of correct predictions made by the model out of all the true instances present in the dataset. A high value of recall signifies that the model has a good ability not to miss the actual, for instance, it will not miss the specific cases of a certain age, gender, or race that the model is looking for.

Recall = True Positives / (True Positives + False Negatives)

D. F1-Score

F1-score gives a comprehensive picture of both precision and recall. Its being the harmonic mean of the two, it is very appropriate to use in situations where the data is unbalanced—something that is usual in crime datasets where some suspect categories are more common than others.

When you get a high F1-score, it shows that the model is excellent at both omitting false positives and getting true positives for suspect predictions.

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

E. Confusion Matrix

A confusion matrix distinguishes the model's predictions into unambiguous classes—true positives, true negatives, false positives, and false negatives—for each suspect attribute. This graphical device allows us to see very clearly the areas in which the model is performing and the areas in which it is making mistakes. For instance, it can indicate if the model frequently confuses certain age groups or accurately tells male and female suspects apart. Thus, the confusion matrix analysis gives us a better understanding of the prediction trends which is not accessible through just looking at the accuracy figures.

VI. Results

"Fig. 1'shows 'The model's prediction of the suspect's age group is characterized by a consistent and balanced result across all categories. The system's overall accuracy of about 69.7

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🎯 Target: AGE GROUP (Overall Accuracy: 0.6970)
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Classification Report (Precision, Recall, F1-Score):

              precision    recall  f1-score   support

  Middle Aged       0.71      0.64      0.67       239
       Senior       0.69      0.73      0.71       239
         Teen       0.68      0.75      0.71       261
  Young Adult       0.72      0.67      0.69       261

     accuracy                           0.70      1000
    macro avg       0.70      0.70      0.70      1000
 weighted avg       0.70      0.70      0.70      1000
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Fig. 1. Classification report for Suspect Age Group showing precision, recall, and F1-scores.

The confusion matrix for the age group classifier indicates that the model is quite steady in its performance across all four classes, with the highest accurate predictions appearing in the groups of Teen and Young Adult. The diagonal elements are pointing to a very good classification for each group, yet there is some overlap—most notably between Middle Aged and Young Adult, which we can assume is due to their being similar in the dataset in terms of behavior. Misclassifications are not much in number and are fairly spread out, which implies that the model has a good grasp of the age-related trends but at times the characteristics of certain groups are so similar that they confuse the model. The matrix in general

shows that the model has a good performance in terms of reliability and consistency when it comes to predicting the age of suspects.

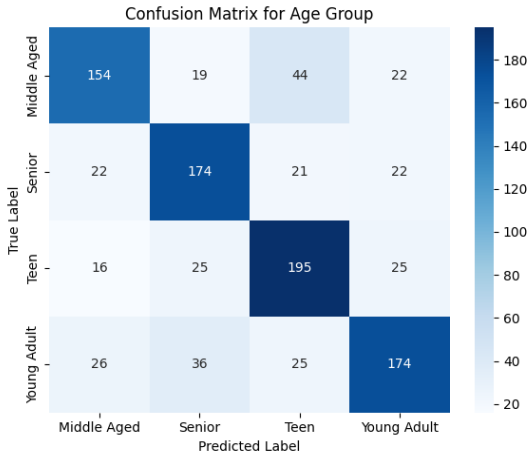

Fig. 2. Confusion Matrix for age group.

"Fig. 3' shows the model's performance in estimating the race of a suspect is rather moderate but, at the same time, pretty consistent, giving an overall of about 65.9

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🎯 Target: SUSPECT RACE (Overall Accuracy: 0.6590)
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Classification Report (Precision, Recall, F1-Score):

              precision    recall  f1-score   support

       Asian       0.65      0.68      0.66       207
       Black       0.63      0.67      0.65       177
    Hispanic       0.65      0.59      0.62       202
       Other       0.64      0.61      0.62       209
       White       0.73      0.75      0.74       205

    accuracy                           0.66      1000
   macro avg       0.66      0.66      0.66      1000
weighted avg       0.66      0.66      0.66      1000
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Fig. 3. Classification report for Suspect Race prediction with an overall accuracy of 65.9.

"Fig. 4'shows moderate performance is the result of race prediction mostly because of the data set's uneven distribution of racial categories. The model can accurately categorize the most frequent race labels, but it cannot cope with the minority groups; hence, certain races get lower precision and recall. The confusion matrix shows that several races are often wrongly classified as more prevalent groups, which is typical for imbalanced datasets. Even though the total accuracy is still good, it may be a time for improving the dataset balance or using more advanced methods like class-weight tuning or oversampling to get better performance in the next iterations.

"Fig.5shows that the model excelled at predicting the suspect's gender with an overall accuracy of 83.7, which is the highest among all the target attributes. Good precision, recall, and F1-scores close to 0.84 are obtained for both

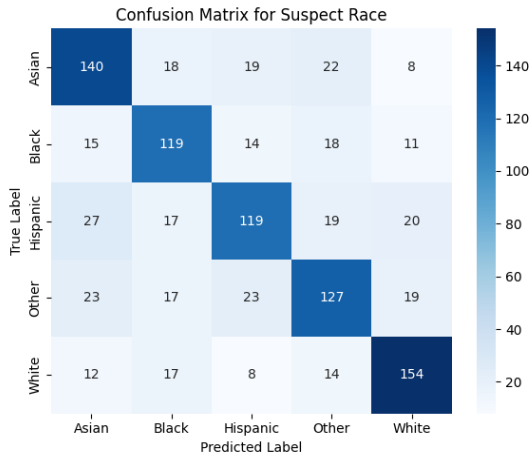

Fig. 4. Confusion matrix and performance summary for Suspect Race prediction, showing moderate accuracy influenced by dataset imbalance across racial categories.

classes—Male and Female—showing that the model has a good balance with respect to correctly identifying true cases and avoiding misclassifications. The minor variations between the two classes indicate that the model has consistently dealt with gender-related patterns throughout the dataset. This strong and stable performance illustrates that gender prediction is one of the most reliable areas of the model.

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🎯 Target: SUSPECT GENDER (Overall Accuracy: 0.8370)
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Classification Report (Precision, Recall, F1-Score):

              precision    recall  f1-score   support

      Female       0.85      0.82      0.84       514
        Male       0.82      0.85      0.84       486

    accuracy                           0.84      1000
   macro avg       0.84      0.84      0.84      1000
weighted avg       0.84      0.84      0.84      1000
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Fig. 5. Classification report for Suspect Gender prediction, showing high and balanced precision, recall, and F1-scores with an overall accuracy of 83.7

In analyzing the confusion matrix, the performance of the model in differentiating male and female suspects is excellent. The majority of the predictions are situated on the diagonal which is indicative of a high number of correct classifications for both categories. The model successfully detects 424 female and 413 male suspects, making a very small number of wrong identifications. The model's error distribution is balanced which implies that the model is able to learn gender-related patterns well and is also reliable in making predictions across both classes. Thus, the matrix presents very high predictive accuracy and indicates that gender is the least challenging attribute for the model to classify.

The feature importance values bring out that the age of the victim is the strongest predictor in the model, contributing

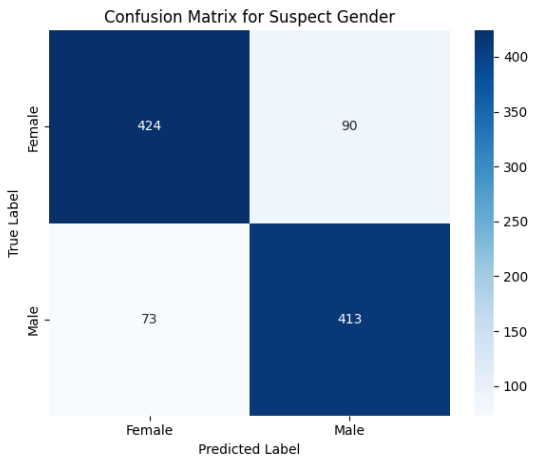

Fig. 6. Confusion Matrix for suspect gender.

much more than any other feature. This indicates that the age patterns in the dataset have a great impact on the formation of various offender characteristics. Other than age, victim gender differences, crime, and modus operandi categories come as the next influencing factors. They have insignificant but still relevant importance scores, thus, making their contributions to the model's predictions. Overall, these findings demonstrate that the model primarily depends on demographic information while at the same time using behavioral and crime-type information to enhance the accuracy of suspect profiling.

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--- 7. FEATURE IMPORTANCE PLOT ---

Top 5 Most Predictive Features:
victim_age                        0.602284
victim_gender_Other               0.035814
modus_operandi_Physical assault   0.035031
victim_gender_Male                0.033614
crime_type_Drug Trafficking       0.032533
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Fig. 7. Feature Importance values.

"Fig.8shows the analysis of feature importance which demonstrates that age of the victim is the most decisive factor in predicting suspect characteristics. Such a large contribution indicates that different crimes are committed according to age-related behavioral patterns to a great extent. There are other features like victim's sex, specific crime types, and certain modus operandi categories that contribute to the model's decisions as well, but their influence is much less. This distribution indicates that although there are several inputs that support the prediction process, the model mainly depends on demographic indicators, especially the victim's age, to create accurate suspect profiles.

Discussion

- The findings of the present research indicate the potential of machine learning to play a significant role in investigative profiling by uncovering correlations in crime-related data and converting them into helpful predictions regarding the features of the offenders, possibly the case of male or female suspects, the range of ages, and the race. The strongest performance of the machine learning model was in predicting the gender of the suspect among the three target attributes: gender, age group, and race, which was coupled with high precision, recall, and overall accuracy. This implies that the gender-related behavioral and situational patterns are well captured by the dataset, enabling the Random Forest model to master these relationships. The corresponding confusion matrix further supports this reliability, as the majority of instances for both male and female suspects were accurately classified.
- The prediction of a suspect's age group was a little bit more than moderately strong and gave constant results in all categories. There were some mixed-up cases between the two nearest age groups, like Middle Aged and Young Adult, but this mixing is considered normal since people in adjacent age brackets often have similar behaviors. Nevertheless, the model still managed to maintain a pretty high accuracy of almost 70, which means that the age-related signals that can be derived from victim information, crime type, and modus operandi have not lost their power in prediction.
- The suspect race predictions, although comparatively weaker, still exhibited reasonable performance patterns. The moderate accuracy can largely be attributed to class imbalance within the dataset, where certain racial categories appear less frequently than others. These imbalances make it more difficult for the model to generalize and accurately identify minority classes. The confusion matrix supported this observation by showing that some race categories were occasionally misclassified into better-represented groups. This indicates that while the model has learned some race-related associations, further refinement of the dataset or the use of balancing techniques may be necessary to improve fairness and accuracy in future iterations.
- The analysis of feature importance showed that victim age is the most important factor in predicting the model and is the one that most influences the pattern of criminal

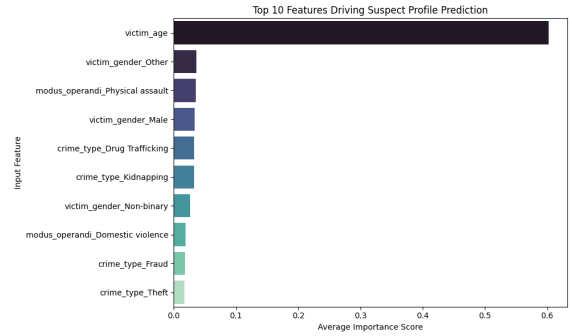

Fig. 8. Top 10 most important features influencing suspect profiling predictions, with victim age identified as the dominant factor.

behavior in the database. Other characteristics such as victim gender, crime type, and modus operandi have an impact on the results but are not as significant. The fact that demographic factors are so dominant implies that the situation of the crime and the characteristics of the victim are very powerful in predicting the traits of the offender.

- There are limitations that the model has despite promising results. The dataset imbalance between races and some crime categories negatively feeds into the predictive performance of the model. For instance, the model cannot capture spatial patterns or contextual insights in narrative reports that are not included in the structured fields, thus limiting its performance. The issue of ethics in the use of the model has not been solved yet, and it is adviced that the model only serves as a decision-support tool and not as the basis for labeling people.
- Overall, the findings show that machine learning can meaningfully support modern investigations. The system offers fast, clear, and reliable insights, helping investigators notice patterns that might otherwise go unnoticed. With improvements in dataset diversity, fairness, and feature representation, the model could become an even more powerful tool for analysis and investigative support.

VII. Future Scope

- The existing framework is already very much oriented towards a data-driven approach to suspect profiling; however, there exist various places where enhancement could be made. The inclusion of more varied and well-distributed crime records in the data set will not only aid in boosting the model's impartiality—particularly in the case of race prediction—but also improve its applicability to a broader scope of circumstances. Future iterations might also include more parameters like geographic location, time-based patterns, or verbal descriptions from police reports, possibly employing NLP and GIS methods to reveal more profound insights into behavior.
- The application of more sophisticated machine learning models, such as gradient boosting techniques or deep neural networks, would probably lead to an even greater prediction accuracy. Moreover, the use of explainable AI tools like SHAP and LIME in conjunction with the model can be an additional support for transparency and can assist the researchers in comprehending the process of prediction making more thoroughly.
- One can see the system as having an extended application to an interactive dashboard or web-based tool capable of online investigations. From the very start of its development, the system will have to deal with ethical issues, such as the need to eliminate bias and to impose responsible usage.

VIII. Conclusion

- This research proves the success of a multi-output machine learning approach for assigning main suspect traits to incident-level crime data. A classifier based on Random Forest and an organized preprocessing pipeline have been combined here. Thus, the system has succeeded in making concurrent forecasts for suspect gender, age group, and race. The highest accuracy among the three targets was reached in gender prediction, then age group and race followed. These findings point out the fact that there are significant behavioral and demographic patterns in crime datasets which, if properly handled, can aid the investigators in making a decision.
- The examination of feature importance reveals that the demographics of the victims, especially their age, were the major factors affecting the model's predictions. Race prediction, however, was a weak point in the model, and this was due to class imbalance and insufficient representation in the dataset that had already been used. It is again pointed out that the use of various and impartial datasets is an essential part in the creation of predictive tools for sensitive applications.
- Importantly, the system is not meant to take over human decisions but rather to be a supporting analytical tool that could help investigators to eliminate the possible traits of the suspect. By providing the model with stable and factual insights, there will be a possibility of lessening the manual work, revealing the concealed patterns, and giving more clarity about the situation in the initial moments of the investigation.
- All in all, this project has laid a solid groundwork for the use of machine learning in suspect profiling. If the areas of dataset diversity, algorithmic fairness, and feature richness are to be the next focus of development, the systems could then be considered as dependable tools in the tech-savvy, modern police force.

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